Fueling the Brain

The brain is an energy-hungry organ. Unlike other organs in the body, it is fueled solely on glucose, a simple sugar that is quick and easy to metabolize.

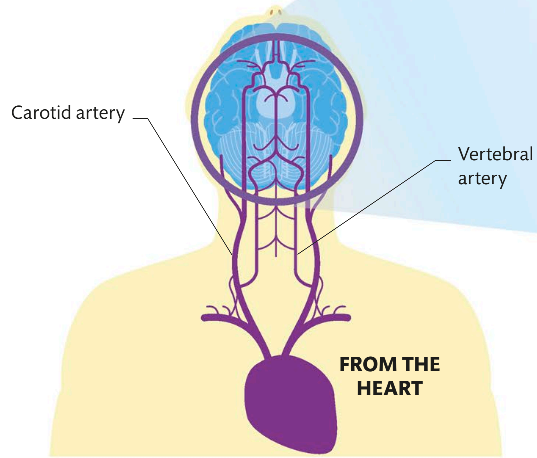
DOES FOCUSED CONCENTRATION USE MORE ENERGY?

The brain never stops working, and the overall energy consumption stays more or less the same 24 hours a day.

Blood supply

The heart supplies blood to the whole body, but around a sixth of its total effort is devoted to sending blood up to the brain. Blood reaches the brain by two main arterial routes. The two carotid arteries, one running up each side of the neck, deliver blood to the front of the brain (and the eyes, face, and scalp). The back of the brain is fed by the vertebral arteries, which weave upward through the spinal column. Deoxygenated blood then accumulates in the cerebral sinuses, which are spaces created by enlarged veins running through the brain. The blood there drains out of the brain and down through the neck via the internal jugular veins.

The vascular system delivers 26 fl oz (750 ml) of blood to the brain every minute, which is equivalent to 1.7 fl oz (50 ml) for every 3.5 oz (100 g) of brain tissue. If that volume drops below about 0.7 fl oz (20 ml), the brain tissue stops working.



Crossing the blood-brain barrier

The blood-brain barrier is a physical and metabolic barrier between the brain and its blood supply. It offers extra protection against infections, which are hard to combat in the brain using the normal immune system, and could make the brain malfunction in dangerous ways. There are six ways that materials can cross the barrier. Other than that, nothing gets in or out.

Paracellular transport Water and water-soluble materials, such as salts and ions (charged atoms or molecules), can cross through small gaps between capillary-wall cells.

Diffusion Cells are surrounded by a fatty membrane, so fat-soluble substances, including oxygen and alcohol, diffuse through the cell.